

90-100

90 x 25. 2009/60

20

5-6

sorting

Algorithm

placement

Le-0

Lec_57

$n \log n$

Heap Sort Algorithm

→ Equation of Heap

$O(n \log n)$

Heapify method

Build Heap method

$O(n)$

Sorting: arrange the things in increasing or decreasing order
→ Bubble, select, In, ~~qu~~, merge, ~~heap~~

Heap sort Algorithm:

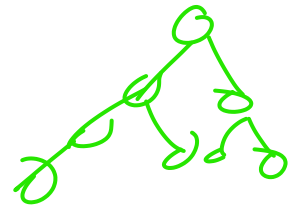
- Given an array , push heap
- Using build heap construct max heap tree , $O(n)$
- Delete one by one element and store them RHS
Continue n - times , $n \log n$
- $TC = n \log n$ [heap sort $TC = n \log n$]
- If all elements are same heap sort will take
 $n + n, O(n)$

→ sort the array using heap sort

① arr[] = [6, 9, 10, 2, 4, 1, 3, 5, 51, 100, 102, 93, 36]

sort in increasing order using heap sort
algo of heap sort: $\rightarrow$ min heap

Step 1: build heap (max or min)
 using given array, with the help of
Build heap method



max
 1st del = 1st max
 2nd del = 2nd max

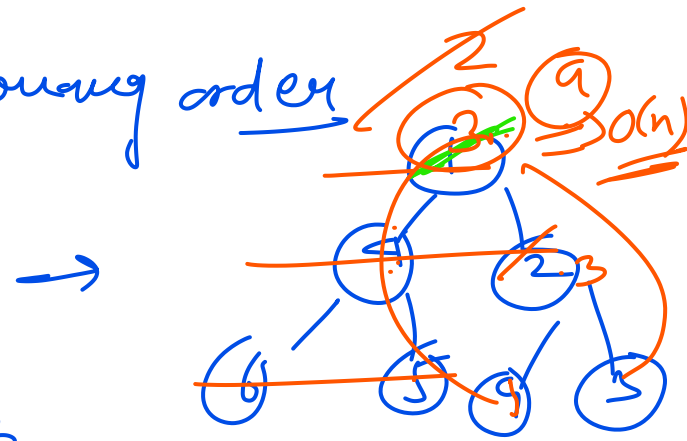
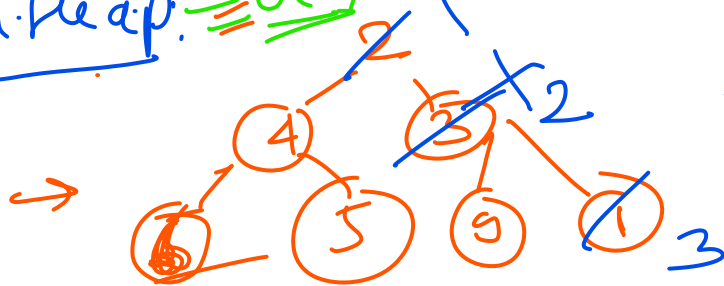
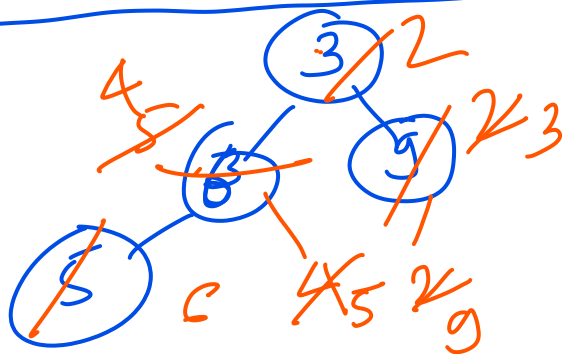
min
 1st del = 1st min
 2nd del = 2nd min

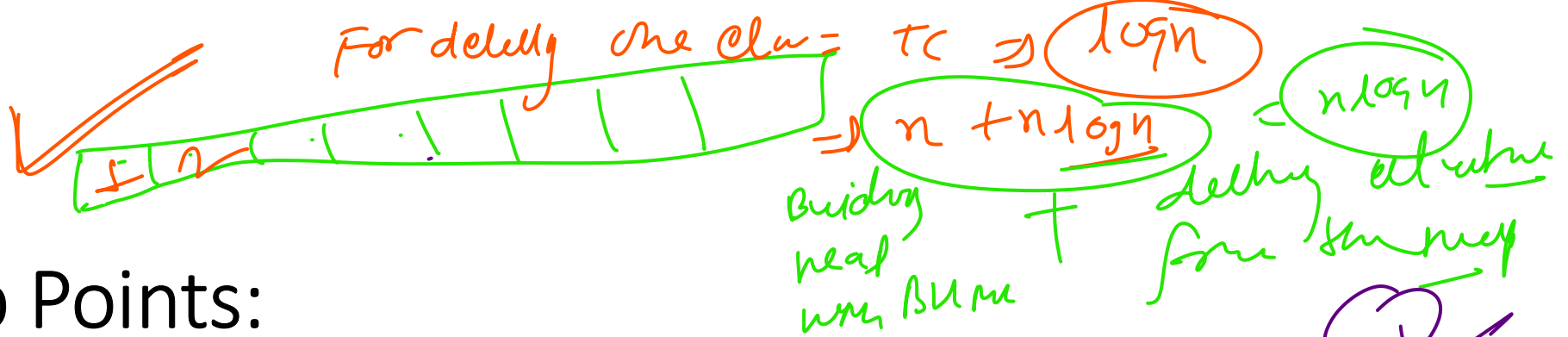
Step 2: delete element from heap
one by one and store it in different
array or at the last of same array.

Ex: [3, 6, 9, 5, 4, 2, 1]

array in increasing order $\rightarrow$

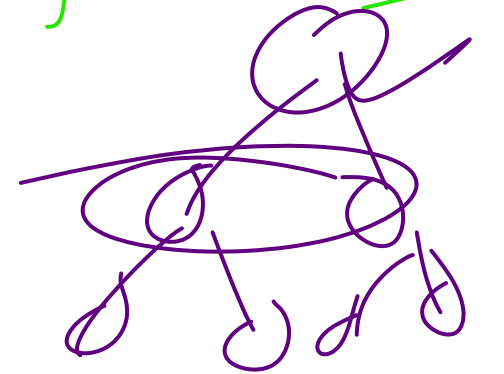
Build heap method min heap = $O(n)$





Imp Points:

- In max heap finding 1st max = $O(1)$
- In max heap finding 2nd max = $O(1)$
- In max heap finding 3rd max = $O(1)$
-
-
- In max heap finding constant max = $O(1)$



1st, 2nd, 3rd, 100th, 10th, 100th

For kth max = $k \log n$

just

In max heap to find minimum Element = $O(n)$



becaus in max heap min element
is present in leaf node

lf = $n/2$ $\Rightarrow$ $O(n/2) = O(n)$

Imp Points:

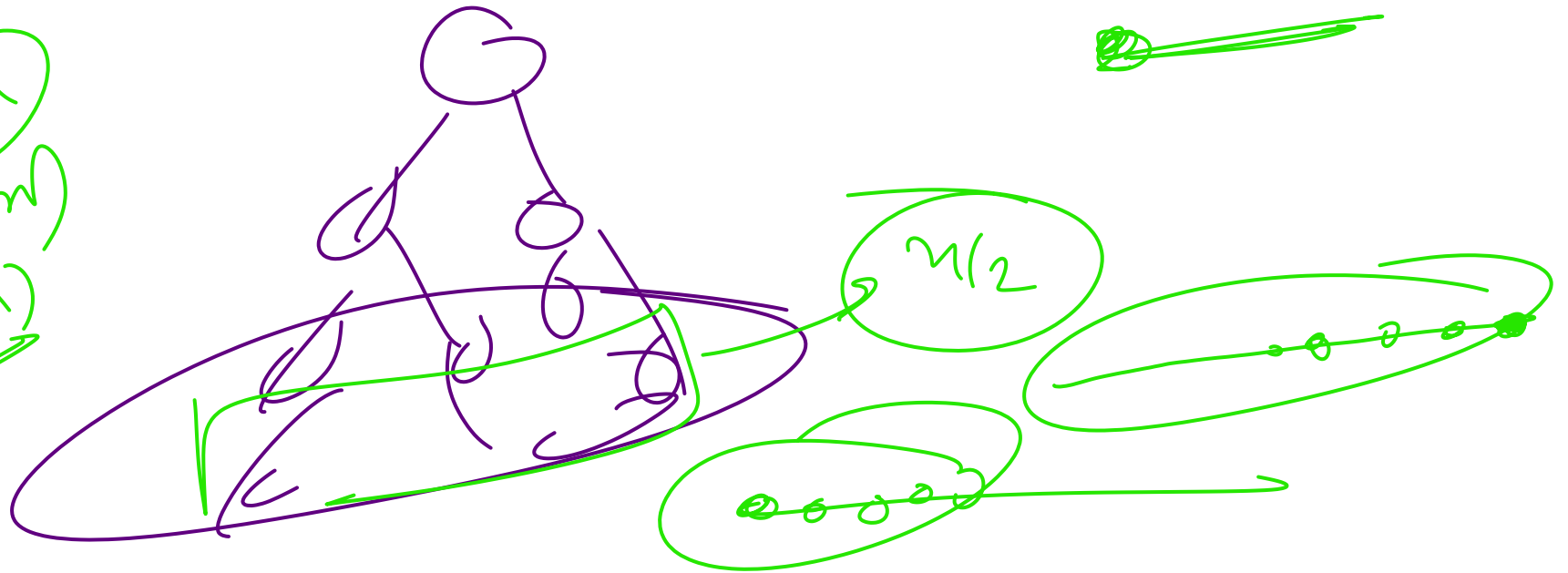
- In max heap deleting 1st max = $O(1)$
- In max heap deleting 2nd max = $O(1)$
- In max heap deleting 3rd max = $O(1)$
- In general In maxheap for finding or deleting kth max = $O(k \log n)$

apply 25 passes of
select

min

In general In maxheap for finding 25th ~~max~~ = $O(25n)$
, $O(n)$

$25 \times n$
 $O(n \log n)$
 $O(n)$



decreasing

Maxheap tree : Increasing order
Minheap tree : decreasing order

- Inplace
- Not stable

(n log n)

outplace $\Rightarrow$ merge sort
heap $\Rightarrow$ heap sort
selection
quick

Thanks

Greedy technique

